Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.



kredibel

## Every number has a source.

An AI finance workspace that checks a small business's own books, built on Gemini, ADK, Cloud Run, Firestore and Firebase.

Team details

| | |
|---|---|
| Team name | Kredibel |
| Team leader | Muhammad Fadly Tanjung |
| Problem statement | Theme 1, BFSI: Intelligent Risk, Fraud and Financial Experiences. Small businesses close their books by hand, so duplicates, edited documents and diverted payments go unnoticed. |

Brief about the idea



# Kredibel reads every number twice, and only counts it when both readings agree

Kredibel is an AI finance workspace for small businesses and their accountants. Upload bank statements, invoices, receipts and the general-ledger export in any order.

It catches duplicates and edited documents before they are counted, reconciles every month, sends the few things AI cannot settle to a short review queue, and shows each month's fraud risk with the evidence and the next steps.

**Any order**

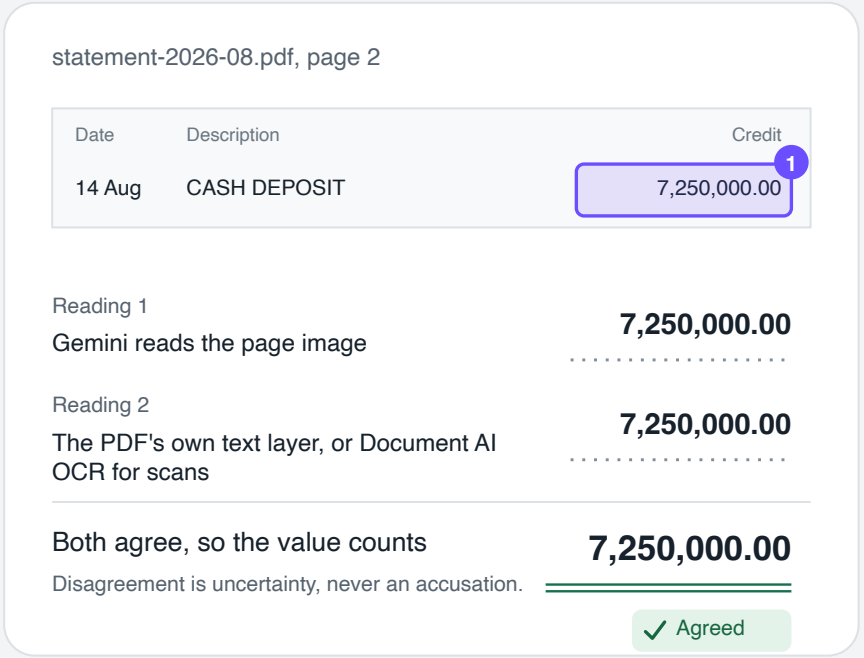
same data, same result

**Page and box**

behind every value

**95%**

of bank lines matched automatically



The problem



# Small businesses close their books by hand, and three kinds of mistakes slip through

## Duplicates

**×2**

one expense, counted twice

The same statement uploaded twice, a supplier invoice scanned again, a receipt photographed by two people. Expenses are counted twice and suppliers paid twice.

## Edited documents

**85.6%**

of fraud and risk leaders name bank statements the document most open to manipulation

Editing a PDF takes minutes. A changed amount or a balance that no longer adds up looks perfectly normal on screen.

## Diverted payments

**USD 2.77 B**

lost to business email compromise in the US in 2024

An invoice paid to a bank account the supplier never used before is how payment diversion works.

Fraud costs organizations about 5% of revenue a year (ACFE). Mistakes surface weeks later, if at all; owners need the evidence, not another black-box score.

Sources: Inscribe 2026 Document Fraud Report (vendor survey); FBI IC3 2024 report; ACFE Occupational Fraud 2024.

Opportunities



# Extraction tools read a document once. Kredibel checks it against itself, the bank and the ledger

| | Typical OCR and expense tools | Kredibel |
|---|---|---|
| Reading | ✕ Reads once and trusts the output | ✓ Reads twice; a value counts only when both readings agree |
| Duplicates | ✕ Counted again | ✓ Exact copies never read; near copies held for a person |
| Edited PDFs | ✕ Not checked | ✓ Running balances, revision history, fonts in amount columns |
| Fraud signal | ✕ A score without reasons | ✓ A policy file sets the level; each finding opens its page |
| Upload order | ✕ Can change the result | ✓ Same data, same result, in any order |

## How it solves the problem

Duplicates are stopped at upload, edits are caught by forensics on grounded values, and diverted payments surface when reconciliation meets a supplier account it has never seen.

Our USP

**Every number has a source.**

Every match, finding and figure opens the page and box it came from, and the AI explains while code decides.

List of features offered by the solution



# Seven checks between an upload and a closed month

**1** Duplicates first — D01 · D02–D06

Exact copies are never read again; a re-scanned invoice or re-photographed receipt is held until a person decides.

**2** Read twice, pay once

Gemini reads each page image; the text layer or Document AI OCR is the witness. Every value keeps its page and box.

**3** Ledger files by code — Once per layout

Code maps columns from header words and value shapes; Gemini only for headers it cannot place; confirmed once per layout.

**4** Forensics — A01 · M03 · M04 · X01 · X08

Running balances, totals, PDF revision history, odd fonts in amount columns, statement continuity, a supplier's new account.

**5** Self-reconciling months — Any upload order

Exact, split, net-of-tax and ledger matches with reasons; only unique, high-scoring matches are automatic.

**6** Short review queue — Every fix audited

Proposed matches, missing documents, possible copies and uncertain values, each with the evidence and only the fixes that make sense.

**7** Explainable risk — Critical · High · Medium · Low

A policy file sets each month's level; Gemini explains it and orders the next steps; a grounding guard checks every number it writes.

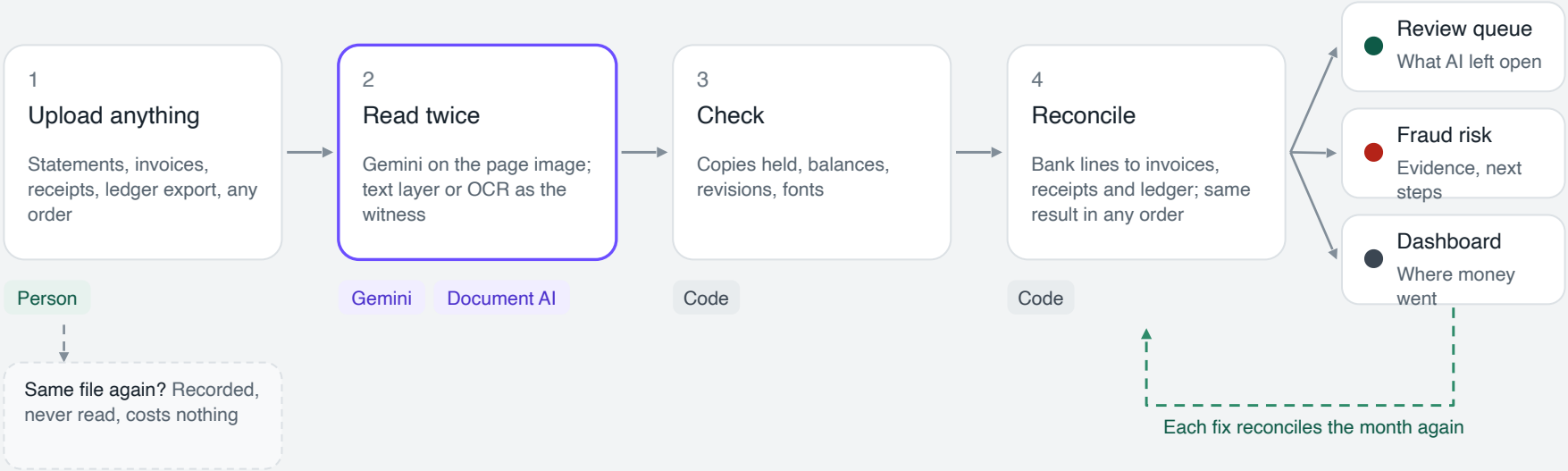
## Around it

Dashboard against what the owner expected, money-flow Sankey and P&L; Transactions with each line's IFRS and US GAAP class; a Finance Assistant that cites its sources; labels people confirm; every run traced; a public demo account and a guided tour.

Process flow diagram

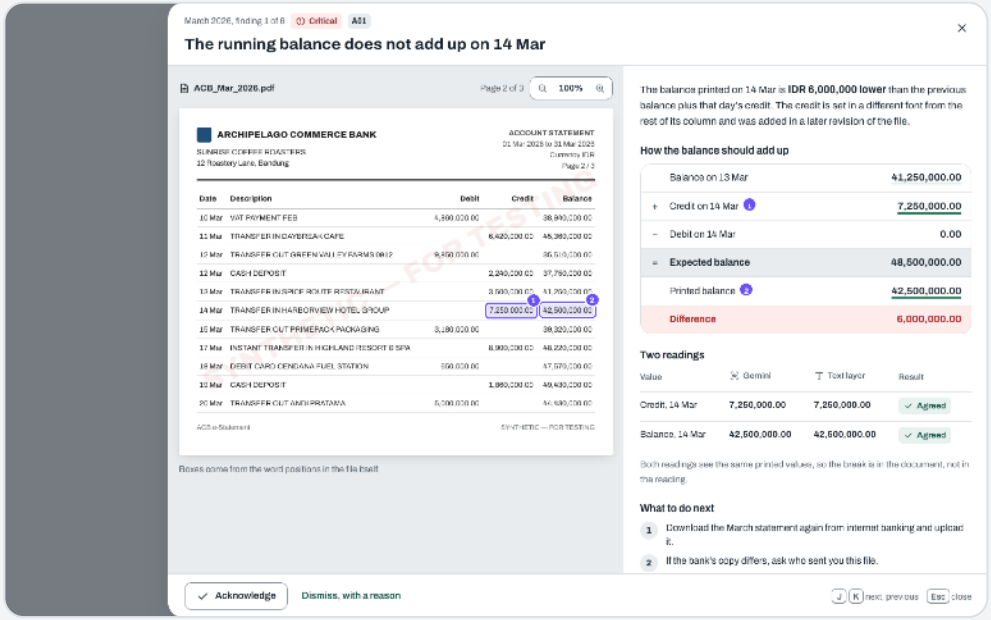


# One upload, from any file to a month you can trust

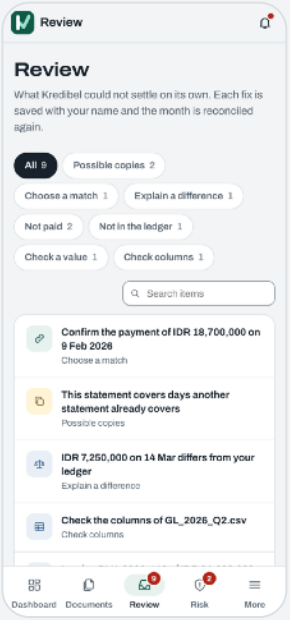



Who does the work Gemini reads and explains Code decides every number Person resolves exceptions

# 43 screens designed first, at phone, tablet and desktop widths




Evidence viewer: the lamp on the edited credit, both readings, the broken balance




Review queue, 390 px

## Evidence first

Every figure is one tap from the page and box it came from.

## Shape before colour

Double, dotted and wavy rules say agreed, one reading, conflict.

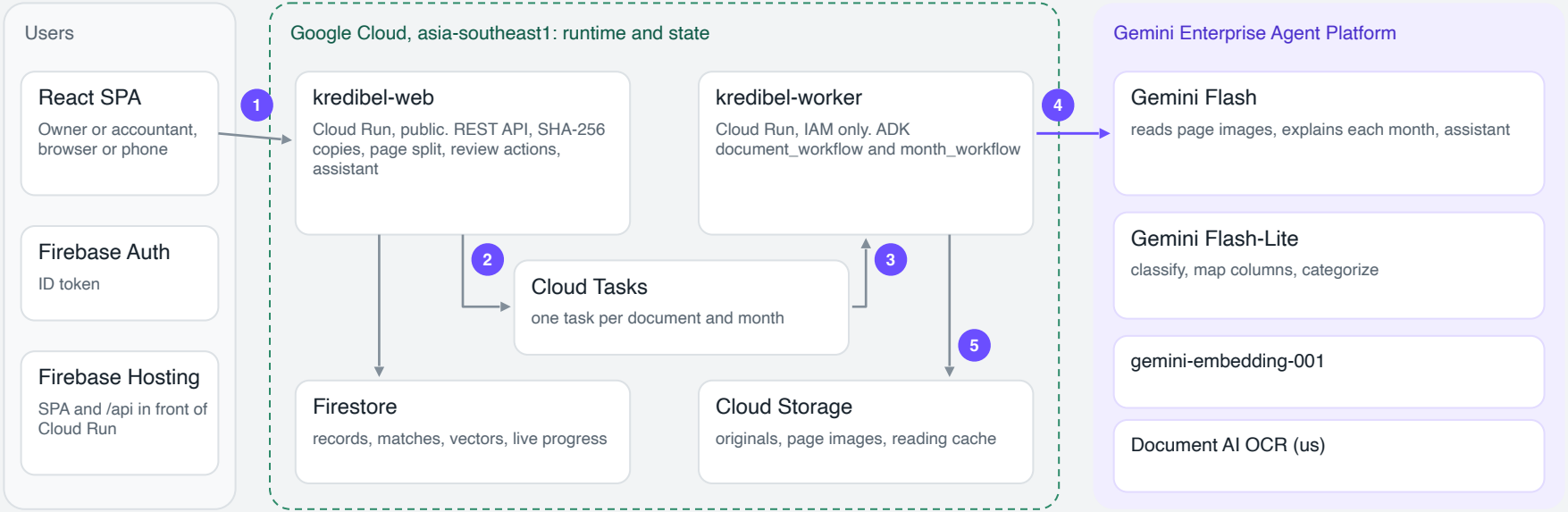
## Works at 360 px

Photograph a receipt or settle a review item on a phone.

Architecture diagram of the proposed solution



# Two ADK workflows on Cloud Run, fed by Cloud Tasks, with Gemini as reader and explainer




1 Upload with ID token
2 Queue the document
3 Run the workflow
4 Read and explain
5 Write results; the app updates live

Delivery and operations: Cloud Build, Artifact Registry, Secret Manager, Cloud Trace (ADK spans), Cloud Logging, Cloud Monitoring with budget alerts

Technologies to be used in the solution



# Google Cloud for every managed service and model; open source only inside our containers

| | |
|---|---|
| Agents | Agent Development Kit 2.x graph workflows, one per document and one per month; traces in Cloud Trace |
| Documents | Gemini on page images, Document AI Enterprise OCR for scans and photos |
| Data | Firestore with vector search, Cloud Storage |
| Delivery | Cloud Build, Artifact Registry, Cloud Scheduler, Cloud Logging, Cloud Monitoring |

| | |
|---|---|
| Models | Gemini Flash and Flash-Lite, gemini-embedding-001, on Gemini Enterprise Agent Platform |
| Runtime | Cloud Run (web and worker), Cloud Tasks, Firebase Hosting |
| Identity | Firebase Authentication (Google or email), IAM-only worker with OIDC, Secret Manager |
| Planned, P1 | Model Armor screening, Agent Platform Sessions and Memory Bank for the assistant |

Open source inside: FastAPI · Pydantic · pikepdf · React 19 · Vite · Tailwind v4 · TanStack Query · d3-sankey · Storybook · Playwright

Model IDs are configuration, never literals in code; prompts are versioned files; every limit lives in a plan contract enforced on the server.

# Gemini reads and explains. Code decides every number

## Where Gemini works

| | |
|---|---|
| Read each page image into values, citing the layout chunk of every value | Flash |
| Classify each document | Flash-Lite |
| Propose ledger columns only when code cannot place a header | Flash-Lite |
| Categorize bank lines in cached batches of 100 | Flash-Lite |
| Explain each month's risk and order the next steps | Flash |
| Answer questions from the books, quoting the pages used | Flash + embeddings |

## Where code decides

- ✓ Grounds every value in the page's own words; the model never marks itself right
- ✓ Duplicates by SHA-256 and by content; held, never deleted
- ✓ Forensics, matching and totals in Decimal; same data, same result
- ✓ Risk levels from a versioned policy file
- ✓ A grounding guard checks each number Gemini writes before you see it

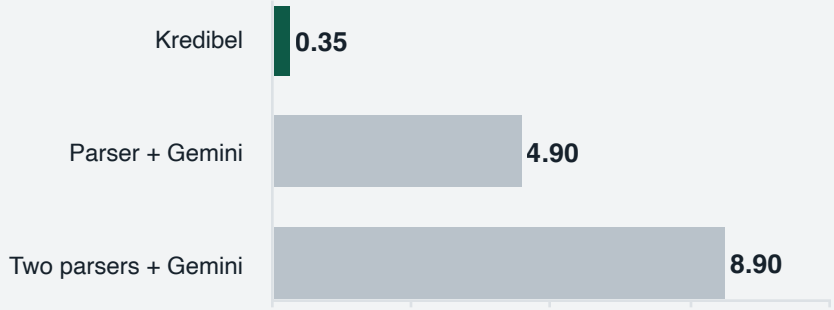
Documents are adversarial. Page text is data, never instructions; structured outputs only; prompts live in versioned files.

Estimated implementation cost



# About USD 0.45 to read and check a half-year of books

Reading the statements and invoices of a half-year (scenario S1), USD




Gemini reads each page image; the free PDF text layer, or Document AI OCR at USD 1.50 per 1,000 pages for scans, is the witness. Exact duplicates cost nothing.

**USD 0.43**

Measured: one full reading of an earlier 80-document demo company (89 pages).

## Unit costs

| | |
|---|---|
| 3-page bank statement | **USD 0.02** |
| Invoice | **USD 0.004** |
| Receipt photo | **USD 0.005** |
| Ledger file, new layout | **USD 0.001** |
| Month run | **USD ≤ 0.005** |
| Assistant question | **USD 0.015** |

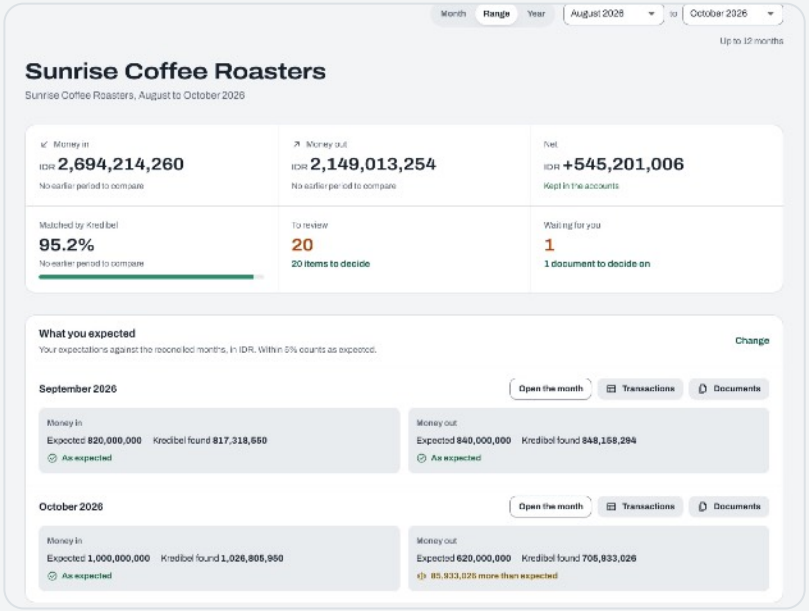
## Running it

Serverless; Cloud Run scales to zero. Firestore for a half-year import ≈ USD 0.01.

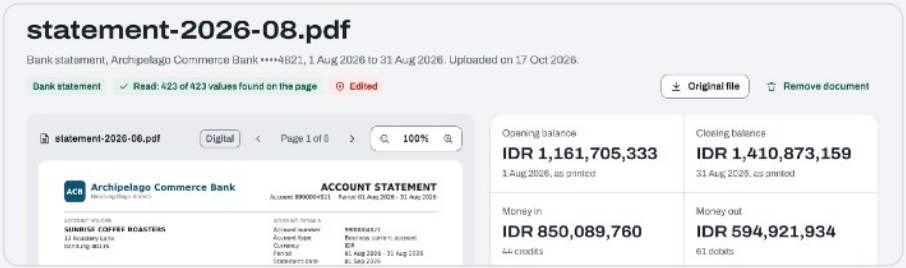
Free trial capped at USD 5 of estimated AI spend per workspace per month.

Expected competition spend ≈ USD 60 to 90.

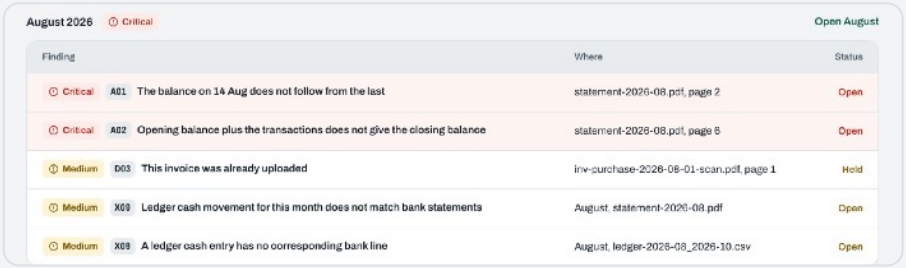
# The working prototype, live on Cloud Run




Dashboard: August to October against what the owner expected




Document: 423 of 423 values found on the page, and the edit flagged




Risk: every finding with its code and its page

# Measured in production and on real documents, gated in CI

**718 / 718**

values grounded on a real 13-page bank statement with 178 lines

**95%**

of bank lines matched automatically in the 300-document demo company (295 of 310)

**14×**

lower reading cost than Form and Invoice Parser for the same documents (USD 0.35 vs 4.90)

Scenario gates, automated tests on synthetic books with known answers

| | | | |
|---|---|---|---|
| S1 | Clean half-year | 85% or more of bank lines matched automatically, none wrong | Pass |
| S2 | Edited March statement | Tampered, with all six expected findings | Pass |
| S3 | Duplicates | Exact copy not read again; four content copies held | Pass |
| S4 | Payment diversion | Supplier's new account (X08) and unknown supplier (R02) | Pass |
| S5 | Reverse upload order | Same matches and figures as S1 | Pass |

Also: 848 backend tests, 217 synthetic-data tests, 265 frontend tests and 216 browser checks against the mocks at three widths. The eval runner with a Gemini judge is next (spec 012).

Additional details and future development



# Next: from checked books to a loan readiness report

## Built and deployed

October 2026

- Reading twice with grounding
- Duplicates and forensics
- Reconciliation in any order
- Risk with explanations
- Review queue, dashboard, P&L, transactions
- Demo company, demo account, guided tour

## Next

P1

- Model Armor screening of every file
- Assistant agent with Memory Bank
- Team roles, invites, month sign-off
- Eval suite and BigQuery reporting

## Later

Production

- Bank feeds and e-invoice validation
- Accounting-tool connectors
- Loan readiness report from verified months
- In-region processing, VPC Service Controls

**Responsible AI**

- ✓ People resolve exceptions
- ✓ Copies held, never deleted
- ✓ No personal attributes
- ✓ Synthetic data in tests

Provide links to your



# Open the demo company and see it for yourself

| | | |
|---|---|---|
| Final product | **https://kredibel.app**<br>Demo sign-in demo@kredibel.app / Kredibel-Demo-2026, or any Google account: explore the demo company, or upload the sample books linked in the README. | |
| GitHub repository | **https://github.com/fadlytanjung/kredibel**<br>Specs, architecture decisions, contracts, design tokens and the code. | |
| Demo video, 3 minutes | Add video link<br>Add the public video link before sending. | |

All documents in the demo are synthetic and marked SYNTHETIC, FOR TESTING.

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# Thank you

